

BIOS 755: Missing Data in Longitudinal Studies

Alexander McLain

March 31, 2026

- ▶ Even in well-controlled situations, missing data invariably occur in longitudinal studies.
- ▶ Subjects can be missed at a particular measurement wave, with the result that these subjects provide data at some, but not all, study time points.
- ▶ Alternatively, subjects who are assessed at a given study time point might only provide responses to a subset of the study variables, again resulting in incomplete data.
- ▶ Finally, subjects might dropout of the study or be lost to follow-up, thus providing no data beyond a specific point in time.

Missing data Mechanism

- ▶ In general, the situation may often be quite complex, with some missingness unrelated to either the observed or unobserved response, some related to the observed, some related to the unobserved, and some to both.
- ▶ A particular pattern of missingness that is common in longitudinal studies is 'dropout' or 'attrition.' This is where an individual is observed from baseline up until a certain point in time, thereafter no more measurements are made.
- ▶ Possible reasons for dropout:
 1. Recovery;
 2. Lack of improvement or failure;
 3. Undesirable side effects;
 4. External reasons unrelated to specific treatment or outcome;
 5. Death

An Example

Schizophrenia Treatment Trial

- ▶ A randomized longitudinal study of haloperidol and risperidone
- ▶ Primary outcome: Positive and Negative Scale for Schizophrenia (PANSS) score at 8 weeks, intermediate outcomes at 1, 2, 4, and 6 weeks

Placebo	$27/88 = 31\%$
Haloperidol	$36/87 = 41\%$
Risperidone (6mg)	$52/86 = 60\%$

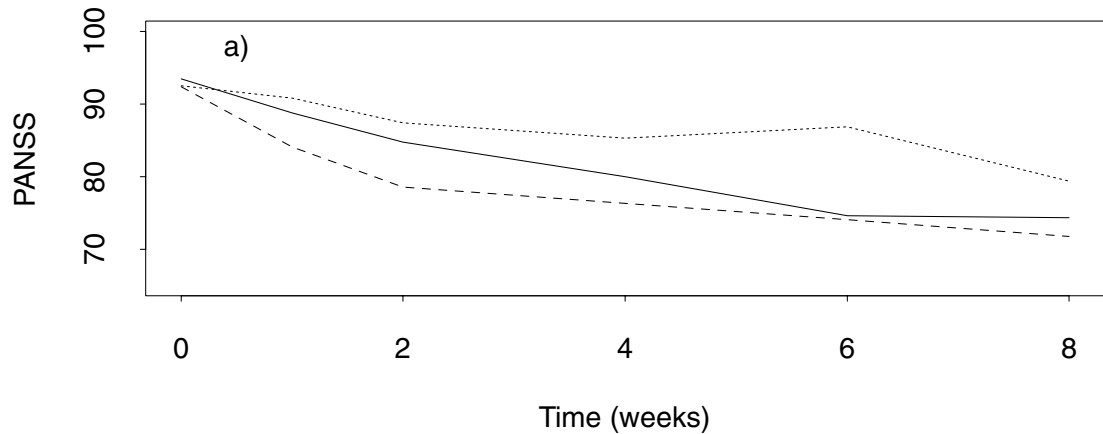
An Example

► Reasons for dropout:

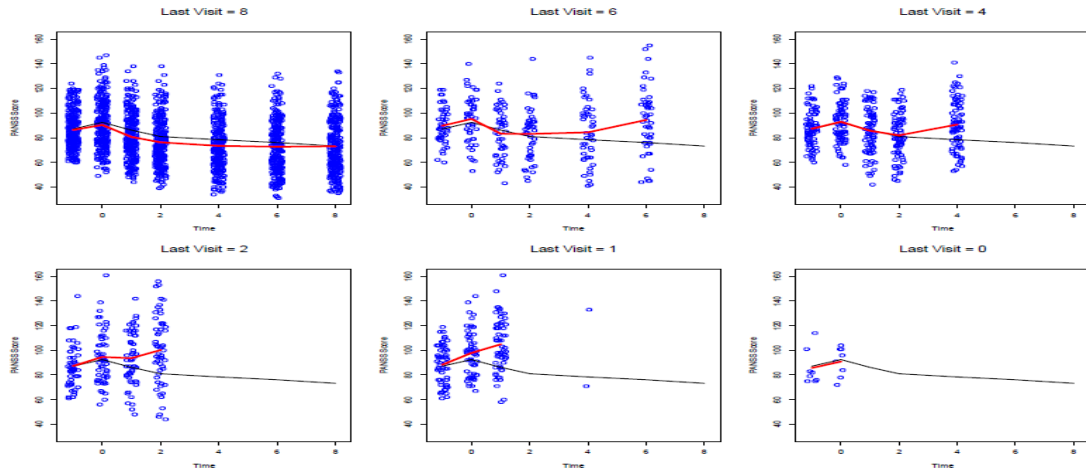
Abnormal lab result	4
Adverse experience	26
Inadequate response	183
Inter-current illness	3
Lost to follow-up	3
Uncooperative	25
Withdrew consent	19
Other	7

► “Inadequate response” is seen as potentially informative.

An Example



An Example



Another Example

Six Cities Study of Air Pollution and Health

- ▶ Longitudinal study designed to characterize lung growth as measured by changes in pulmonary function.
- ▶ A cohort of 13,379 children born on or after 1967 was enrolled in six communities across the U.S.
- ▶ Most children were enrolled in the first or second grade (between the ages of six and seven) and measurements of study participants were obtained annually until graduation from high school or loss to follow-up (1 to 12 measurements were taken).
- ▶ Major reason for late entry or attrition was moving in or out of the school district.

Another Example

Muscatine Coronary Risk Factor (MCRF)

- ▶ This study which had five cohorts of children, initially aged 5–7, 7–9, 9–11, 11–13, and 13–15.
- ▶ Goal: determine whether the risk for obesity increased with age.
- ▶ Measurements were taken in 1977, 1979 and 1981. Less than 40% had complete data.
- ▶ Major reasons for missing data:
 - ▶ failure to obtain consent from parents, and
 - ▶ the child was absent from school on the day of measurement.

Some Terminology

- ▶ Complete Data: The scheduled measurements. This is the outcome vector, $\mathbf{Y}_i$, that would have been recorded if no missing data occurred.
- ▶ Missing Data Indicators: The binary variables, $\mathbf{R}_i = \{R_{i1}, R_{i2}, \dots, R_{in_i}\}$, that indicate whether Y_{ij} was observed, $R_{ij} = 1$, or unobserved, $R_{ij} = 0$. That is,

$$R_{ij} = 1 \quad \text{if subject } i \text{ is observed at time } j$$

$$R_{ij} = 0 \quad \text{if subject } i \text{ is not observed at time } j$$

- ▶ Observed (Response) Data: $\mathbf{Y}_i^O$: Y_{ij} such that $R_{ij} = 1$.
- ▶ Missing Data: $\mathbf{Y}_i^M$: Y_{ij} such that $R_{ij} = 0$.

Missing Data Patterns and Mechanisms

- ▶ Consider the joint distribution of $\mathbf{Y}$ and $\mathbf{R}$ (both are random variables).
- ▶ The pattern of missingness concerns the distribution of $\mathbf{R}$.
- ▶ Mechanism concerns how the distributions of $\mathbf{R}$ and $\mathbf{Y}$ relate to one another.

Group	Y1	Y2	Y3
1	X	X	X
2	X	X	.
3	X	.	X
4	X	.	.
5	.	X	X
6	.	X	.
7	.	.	X
8	.	.	.

Types of Missing Data

- ▶ We will discuss three different types of missingness mechanisms:
 1. **Missing completely at random (MCAR),**
 2. **Covariate dependent missingness,**
 3. **Missing baseline covariates,**
 4. **Missing at random (MAR), and**
 5. **Missing not at random (MNAR).**

Types of Missing Data

- ▶ We will discuss three different types of missingness mechanisms:
 1. **Missing completely at random (MCAR),**
 2. **Covariate dependent missingness,**
 3. **Missing baseline covariates,**
 4. **Missing at random (MAR), and**
 5. **Missing not at random (MNAR).**
- ▶ To formulate different missing data mechanisms suppose we only observe:
 - ▶ $\mathbf{Y}_i = \{Y_{i1}, Y_{i2}\}$ where Y_{i1} is always observed and Y_{i2} is sometime missing.
 - ▶ Covariate data $\mathbf{X}_i$.
 - ▶ $R_{i1} = 1$ for all individuals, R_{i2} can be either a 1 or 0.

Missing Completely at Random (MCAR)

- ▶ Data are said to be MCAR if

$$\Pr(R_{i2} = 1 \mid \mathbf{Y}_i, \mathbf{X}_i) = \Pr(R_{i2} = 1)$$

- ▶ This implies

$$E(Y_{i2} \mid R_{i2} = 1, \mathbf{X}_i) = E(Y_{i2} \mid \mathbf{X}_i)$$

- ▶ Interpretation:

- ▶ Missingness is unrelated to both observed covariates and outcomes
- ▶ The observed data are a random subset of the full data

- ▶ MCAR could occur if:

- ▶ data were lost due to administrative error,
- ▶ missingness is completely unrelated to patient characteristics or outcomes

Covariate-Dependent Missingness

- ▶ Data are covariate-dependent if

$$\Pr(R_{i2} = 1 \mid \mathbf{Y}_i, \mathbf{X}_i) = \Pr(R_{i2} = 1 \mid \mathbf{X}_i)$$

- ▶ This implies

$$E(Y_{i2} \mid R_{i2} = 1, \mathbf{X}_i) = E(Y_{i2} \mid \mathbf{X}_i)$$

- ▶ Interpretation:

- ▶ Missingness depends on observed covariates $\mathbf{X}_i$
- ▶ Missingness does not depend on the outcome Y_{i2} (given $\mathbf{X}_i$)

- ▶ Examples:

- ▶ dropout related to observed side effects or baseline severity score
- ▶ follow-up depends on demographic or clinical characteristics

Missing Baseline Covariates

- ▶ Consider missingness in a baseline covariate X_i
- ▶ If X_i is included in the analysis model, subjects with missing X_i are typically excluded
 - ▶ (complete-case analysis)
- ▶ Key difference from missing outcomes:
 - ▶ Subjects with missing outcomes still contribute partial information
 - ▶ Subjects with missing X_i contribute no information
- ▶ This can change the target population
- ▶ Multiple imputation can be used to include these subjects

Missing Baseline Covariates: Ignorability

- ▶ Let $R_i^X = 1$ indicate X_i is observed
- ▶ To ignore missingness in X_i , require:

$$\Pr(R_i^X = 1 \mid Y_i, X_i, Z_i) = \Pr(R_i^X = 1 \mid Z_i)$$

where Z_i are fully observed covariates

- ▶ Interpretation:
 - ▶ Missingness may depend on observed data Z_i
 - ▶ Missingness does not depend on X_i or Y_i (given Z_i)
- ▶ Under this assumption:
 - ▶ Complete-case analysis yields valid inference (conditional on Z_i)
 - ▶ Otherwise, estimates may be biased

Missing at Random (MAR)

- ▶ Missing at random (MAR) if

$$\Pr(R_{i2} = 1 \mid Y_{i1}, Y_{i2}, \mathbf{X}_i) = \Pr(R_{i2} = 1 \mid Y_{i1}, \mathbf{X}_i)$$

- ▶ The probability of missingness depends on observed outcomes (e.g., Y_{i1}), but not on the unobserved outcome Y_{i2} .
- ▶ MAR could occur if:
 - ▶ dropout depends on previously observed outcomes,
 - ▶ e.g., subjects with worsening symptoms are more likely to drop out, but this is predictable from Y_{i1} .
- ▶ Important distinction:
 - ▶ With missing outcomes, subjects still contribute partial information
 - ▶ With missing baseline covariates, subjects are often excluded entirely

MAR: Implications

- ▶ Under MAR,

$$E(Y_{i2} \mid R_{i2} = 1, \mathbf{X}_i) \neq E(Y_{i2} \mid \mathbf{X}_i) \quad (\text{in general})$$

- ▶ In general,

$$\Pr(\mathbf{R}_i \mid \mathbf{Y}_i^O, \mathbf{Y}_i^M, \mathbf{X}_i) = \Pr(\mathbf{R}_i \mid \mathbf{Y}_i^O, \mathbf{X}_i)$$

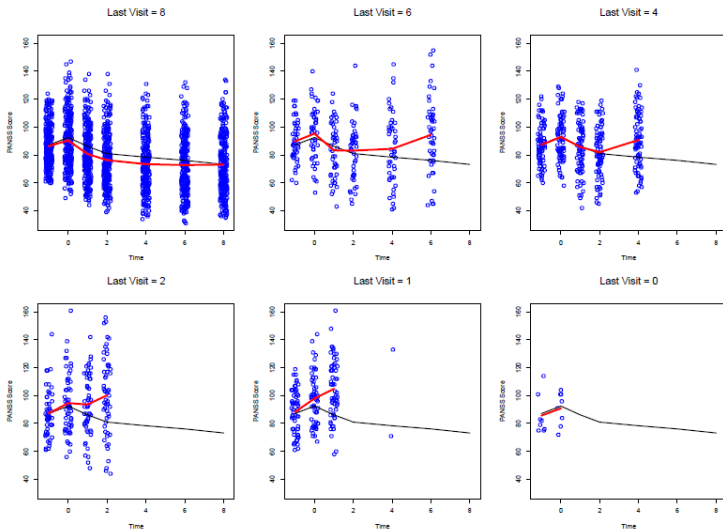
- ▶ Implications:

- ▶ The observed data are not a random sample of the full data
- ▶ However, conditional on observed data $\mathbf{Y}_i^O$, the distribution of missing values $\mathbf{Y}_i^M$ is the same as for observed units

- ▶ Valid inference typically requires:

- ▶ modeling the missingness (e.g., IPW), or
- ▶ modeling the outcome (e.g., imputation, likelihood-based methods)

An Example



MNAR

Missing not at random (MNAR)

- ▶ Also referred to as “nonignorable” missingness or “informative” dropout
- ▶ The data are MNAR if

$$\Pr(R_{i2} = 1 | Y_{i1}, Y_{i2}, \mathbf{X}_i) = \Pr(R_{i2} = 1 | Y_{i1}, Y_{i2}, \mathbf{X}_i)$$

or

$$\Pr(\mathbf{R}_i | \mathbf{Y}_i^O, \mathbf{Y}_i^M, \mathbf{X}_i) = \Pr(\mathbf{R}_i | \mathbf{Y}_i^O, \mathbf{Y}_i^M, \mathbf{X}_i).$$

- ▶ Examples:

1. In the MCRF study, if parent of obese children were more or less likely to sign the consent form.
2. In the Schizophrenia study, subjects dropped out because PANSS increased.

Example of MCAR

- ▶ Lavange and Helms (1983) analyzed data from a longitudinal study of lung function conducted on 72 children age 3 to 12 years;
- ▶ A measure of maximum expiratory flow rate was measured annually, and differences in the resulting curve were related to between-subject covariates such as Race and Gender.
- ▶ The number of actual measurements recorded on each child ranged from 1 to 8, with an average of 4.2 per child.
- ▶ Some values were missing because the child was older than age 3 at the start of the study or younger than 12 at the end of the study.
- ▶ This mechanism depends on cohort, but plausibly does not depend on outcomes if cohort is a covariate in the model.

Example of MAR

- ▶ Murray and Findlay (1988) studied the effect of two drugs in a large multi-center clinical trial.
- ▶ The drugs were antihypertensive agents for patients with mild to moderate hypertension.
- ▶ Diastolic BP was the outcome of interest.
- ▶ The study lasted 12 weeks with measurements at 0, 2, 4, 8, and 12 weeks.
- ▶ The protocol stated that patients with DBP exceeding 110 mmHg at either the 4- or 8-week visit should “jump” to an open follow-up phase (i.e., drop-out).

Example of MNAR

- ▶ Diggle and Kenward (1994) analyzed a longitudinal milk protein trial discussed by Verbyla and Cullis (1990).
- ▶ Cows were randomly allocated to one of three diets: barley, mixed barley-lupins, or lupins, and the assayed protein content of milk samples was taken weekly for a period of 20 weeks.
- ▶ Drop-out corresponds to cows that stopped producing milk before the end of the experiment.